

1. Consideremos la función $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ dada por $f(x,y) = \begin{cases} \frac{\sin(x)}{|x|+|y|} & \text{si } (x,y) \neq (0,0) \\ 0 & \text{si } (x,y) = (0,0) \end{cases}$.

(a) Mostrar que f no es diferenciable en $(0,0)$.

$$f \text{ dif} \Rightarrow \begin{cases} f \text{ cont} \checkmark \\ f \text{ deriv} \checkmark \\ f'(A, \vec{r}) = \nabla f(A) \cdot \vec{r} \checkmark \end{cases} \quad \begin{matrix} A \Rightarrow B \\ -B \Rightarrow -A \end{matrix}$$

$$f(0,0) = 0$$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} \frac{\sin x}{|x|+|y|}$$

$$\text{si } x=0 \rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{0}{|y|} = 0$$

$$x \neq 0 \rightarrow \lim_{(x,y) \rightarrow (0,0)} \frac{\sin x}{|x|+|y|}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x}{|x|+|y|}$$

$$y = mx \rightarrow \lim_{x \rightarrow 0} \frac{x}{|x|+|mx|} = \lim_{x \rightarrow 0} \frac{x}{|x|(1+|m|)}$$

$$\begin{matrix} x \rightarrow 0^+ & \rightarrow & \frac{1}{1+|m|} \\ x \rightarrow 0^- & \rightarrow & -\frac{1}{1+|m|} \end{matrix} \quad \left. \begin{matrix} \text{NO} \\ \text{SON} \\ = \end{matrix} \right\}$$

$$\lim_{x \rightarrow 0} \left(\lim_{y \rightarrow 0} \frac{x}{|x|+|y|} \right) = \lim_{x \rightarrow 0} \frac{x}{|x|} = \begin{matrix} x \rightarrow 0^+ & \rightarrow & 1 \\ x \rightarrow 0^- & \rightarrow & -1 \end{matrix}$$

$$\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow 0} \frac{x}{|x|+|y|} \right) = \lim_{y \rightarrow 0} 0 = 0$$

$\Rightarrow f$ NO CONT. es.

$\Rightarrow f$ NO es dif.

1. Consideremos la función $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ dada por $f(x, y) = \begin{cases} \frac{\sin(x)}{|x| + |y|} & \text{si } (x, y) \neq (0, 0) \\ 0 & \text{si } (x, y) = (0, 0) \end{cases}$.

(a) Mostrar que f no es diferenciable en $(0, 0)$.

(b) Mostrar que la función $g(x, y) = y^2 f(x, y)$ es diferenciable en $(0, 0)$.

$$g(x, y) = \begin{cases} \frac{y^2 \cdot \sin x}{|x| + |y|} & \text{si } (x, y) \neq (0, 0) \\ 0 & \text{si } (x, y) = (0, 0) \end{cases}$$

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{y^2 \cdot \sin x}{|x| + |y|} \quad \text{si } x=0 \rightarrow 0$$

$$\lim_{(x, y) \rightarrow (0, 0)} \frac{y^2 \cdot \sin x}{|x| + |y|} = \lim_{(x, y) \rightarrow (0, 0)} \frac{y^2 \cdot x}{|x| + |y|}$$

$$0 \leq |x| \leq |x| + |y|$$

$$y^2 = |y| \cdot |y|$$

$$0 \leq \frac{|x|}{|x| + |y|} \leq 1$$

$$-1 \leq \frac{x}{|x| + |y|} \leq 1$$

$$0 \leq \frac{|y|}{|x| + |y|} \leq 1$$

$$\lim_{(x, y) \rightarrow (0, 0)} g(x, y) = 0 \rightarrow g(0, 0) \text{ cont.}$$

$$\lim_{h \rightarrow 0} \frac{g((0,0) + h(a,b)) - g(0,0)}{h}$$

$$\lim_{h \rightarrow 0} \frac{g(ha, hb)}{h}$$

$$\vec{v} = (a, b) \\ a^2 + b^2 = 1$$

$$\lim_{h \rightarrow 0} \frac{h^2 b^2 \cdot \sin(ha)}{|ha| + |hb|} \cdot \frac{1}{h} \xrightarrow{a=0} = 0$$

$$\begin{aligned} & \left. \begin{array}{l} a \neq 0 \\ \downarrow \end{array} \right\} \lim_{h \rightarrow 0} \frac{hb^2 \cdot \sin(ha)}{|h|(|a| + |b|)} \cdot \frac{ha}{ha} = \lim_{h \rightarrow 0} \frac{h^2 ab^2}{|h|(|a| + |b|)} \end{aligned}$$

$$= \lim_{h \rightarrow 0} \frac{|h| \cdot |h| \cdot ab^2}{|h|(|a| + |b|)} = 0 \quad \text{by deriv}$$

$$f'_x(A) = f'(A, \hat{i}) = 0$$

$$f'_y(A) = f'(A, \hat{j}) = 0$$

$$\nabla f(0,0) = (0,0)$$

$$\nabla f(0,0) \cdot \vec{v} = 0$$

$$f'((0,0), \vec{v}) = 0?$$

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{f(x,y) - [f(x_0,y_0) + f'_x(x_0,y_0)(x-x_0) + f'_y(x_0,y_0)(y-y_0)]}{\sqrt{(x-x_0)^2 + (y-y_0)^2}} = 0$$

$$\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{f(x,y) - \left[\cancel{f(x_0,y_0)} + \cancel{f'_x(x_0,y_0)}(x-x_0) + \cancel{f'_y(x_0,y_0)}(y-y_0) \right]}{\sqrt{(x-x_0)^2 + (y-y_0)^2}} = 0$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{g(x,y)}{\sqrt{x^2 + y^2}}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{y^2 \cdot \cancel{\sin x}}{|x| + |y|} \cdot \frac{1}{\sqrt{x^2 + y^2}} \cdot \frac{\sqrt{x^2 + y^2}}{\sqrt{x^2 + y^2}}$$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{y^2}{x^2 + y^2} \cdot \frac{\cancel{\sin x} \cdot \sqrt{x^2 + y^2}}{|x| + |y|} \quad \xrightarrow{x=0} 0$$

$$\downarrow x \neq 0 \quad \lim_{(x,y) \rightarrow (0,0)} \frac{y^2}{x^2 + y^2} \cdot \frac{\cancel{\sin x} \cdot \sqrt{x^2 + y^2}}{|x| + |y|} \cdot \frac{\cancel{x}}{\cancel{x}}$$

$$\lim_{(x,y) \rightarrow (0,0)} \left(\frac{y^2}{x^2 + y^2} \cdot \frac{\sqrt{x^2 + y^2} \cdot x}{|x| + |y|} \right) = 0$$

acot acot

$$\boxed{\frac{x}{\sqrt{x^2 + y^2}}} \text{ acot.}$$

$$0 \leq x^2 \leq x^2 + y^2$$

$$0 \leq \frac{x^2}{x^2 + y^2} \leq 1$$

$$0 \leq \frac{|x|}{\sqrt{x^2 + y^2}} \leq 1$$

$$-1 \leq \frac{x}{\sqrt{x^2 + y^2}} \leq 1$$

- Ejercicio 4. Considere la función

$$f(x, y) = \begin{cases} \frac{x^2 \sin y}{x^2 + 2y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Estudie tanto la existencia de las derivadas direccionales como la diferenciabilidad de f en $(0, 0)$.

$$\lim_{h \rightarrow 0} \frac{f(ha, hb)}{h}$$

$$\lim_{h \rightarrow 0} \frac{h^2 a^2 \cdot \sin(hb)}{h^2 a^2 + 2h^2 b^2} \cdot \frac{1}{h}$$

$$b \neq 0 \rightarrow \lim_{h \rightarrow 0} \frac{a^2 \cdot \sin(hb)}{a^2 + 2b^2} \cdot \frac{1 \cdot b}{hb}$$

$$= \frac{a^2 \cdot b}{a^2 + 2b^2} \rightarrow \nexists \forall (a, b) \text{ con } a^2 + b^2 = 1$$

$$\nabla f(0, 0) = (0, 0)$$

$$\nabla f(0, 0) \cdot \vec{r} = (0, 0) \cdot (a, b) = 0 \neq f'(0, 0) \cdot \vec{r}$$

↳ No es dif.

$$\nabla f(A) = (C_1, C_2)$$

$$C_1 \cdot a + C_2 \cdot b$$

22. Dada $y = f(x, z)$ definida implícitamente por $\overbrace{xy + xz + \ln(1 + yz) - 4}^{F(x, y, z)} = 0$ en un entorno de $(x_0, z_0) = (2, 0)$, calcule aproximadamente $f(1.98, 0.03)$ mediante Taylor de 1º orden.

$$p(x, z) = f(x_0, z_0) + f'_x(x_0, z_0)(x - x_0) + f'_z(x_0, z_0)(z - z_0) \approx f(x, z)$$

$$f(\overset{x}{2}, \overset{z}{0}) = y_0 \quad 2y_0 - 4 = 0$$

$$y_0 = 2$$

$$f'_x(2, 0) = \frac{\partial y(2, 0)}{\partial x} = -\frac{F'_x(2, 2, 0)}{F'_{y_0}(2, 2, 0)} = -\frac{2}{2} = -1$$

$$f'_z(2, 0) = \frac{\partial y(2, 0)}{\partial z} = -\frac{F'_z(2, 2, 0)}{F'_{y_0}(2, 2, 0)} = -\frac{4}{2} = -2$$

$$\overbrace{xy + xz + \ln(1 + yz) - 4}^{F(x, y, z)} =$$

$$F'_x = y + z \xrightarrow{(2, 2, 0)} 2$$

$$F'_z = x + \frac{y}{1 + yz} = 4$$

$$F'_y = x + \frac{z}{1 + yz} = 2$$

$$p(x, y) = 2 - (x - 2) - 2 \cdot z \approx f(x, y)$$

$$f(1.98, 0.03) \approx 2 - (1.98 - 2) - 2 \cdot 0.03 = 1.96$$

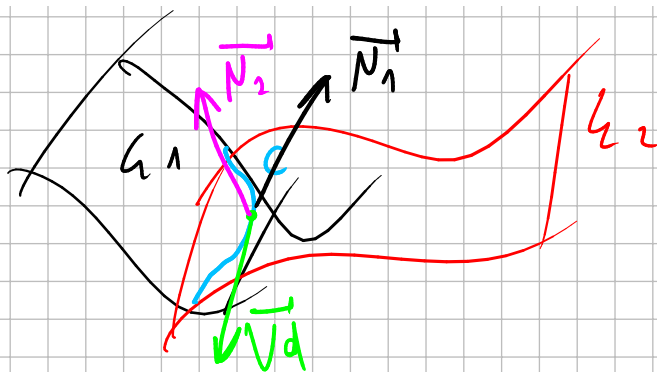
$$f(1.98, 0.03) \approx 1.96$$

10. Dada la curva C definida como intersección de las superficies Σ_1 y Σ_2 cuyas ecuaciones son:

$$\Sigma_1: z = x^2 + y + 1 \text{ con } (x, y) \in \mathbb{R}^2$$

$$\Sigma_2: \vec{X} = (u, u^2, v) \text{ con } (u, v) \in \mathbb{R}^2$$

analice si la recta tangente a C en $(2, y_0, z_0)$ tiene algún punto en común con el plano xz . $y=0$



$$z_0 = 4 + y_0 + 1$$

$$z_0 = 5 + y_0$$

$$(u_0, u_0^2, v_0) = (2, y_0, z_0)$$

$$u_0 = 2$$

$$u_0^2 = 4 = y_0 \quad z_0 = 9$$

$$\Sigma_1: 0 = x^2 + y + 1 - z$$

$$\nabla F = (2x, 1, -1)$$

$$P = (2, 4, 9)$$

$$\vec{N}_1(2, 4, 9) = (4, 1, -1)$$

$$v_0 = 9$$

$$\vec{X}'_u = (1, 2u, 0) = (1, 4, 0)$$

$$\vec{X}'_v = (0, 0, 1) = (0, 0, 1)$$

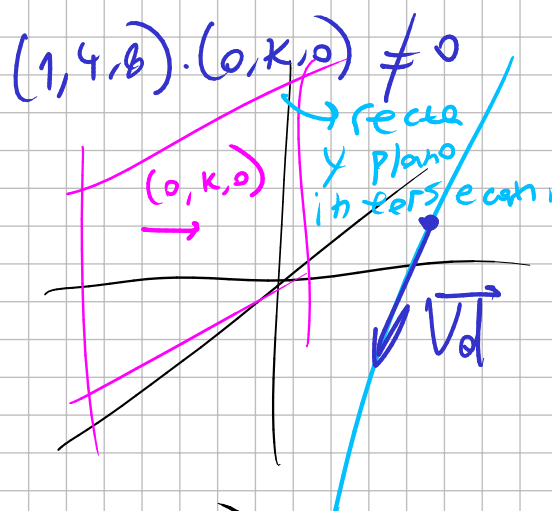
$$(4, -1, 0) = \vec{N}_2(2, 4, 9)$$

$$\Sigma_2: \begin{cases} x = u \\ y = u^2 \\ z = v \end{cases} \quad \begin{cases} y = x^2 \\ y - x^2 = 0 \end{cases}$$

$$\nabla F = (-2x, 1, 0) = (-4, 1, 0)$$

$$\vec{V}_{T_C}(2, 4, 9) = (4, 1, -1)$$

$$\vec{V}_{T_C}(2, 4, 9) = \frac{(4, 1, -1) \times (4, -1, 0)}{(-1, -4, -8)} \rightarrow (1, 4, 8)$$



$$\lambda(1, 4, 8) + (2, 4, 9)$$

$$(\lambda + 2, 4\lambda + 4, 8\lambda + 9)$$

$$= 0$$

$$\lambda = -1$$

$$A = (1, 0, 1)$$